AP BIOLOGY TOPIC 1.5

Lipids: Fatty Acids, Triglycerides, Phospholipids & Steroids

UNIT 1: CHEMISTRY OF LIFE (8–11% EXAM WEIGHT)

SYI-1.B | SATURATION & FLUIDITY | AMPHIPATHIC BILAYER

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 1.5 (LIPIDS: FATTY ACIDS, TRIGLYCERIDES, PHOSPHOLIPIDS & STEROIDS)

CED Core Vocabulary

Defining Characteristic — Grouped together NOT by repeating monomer polymers, but because they are mostly nonpolar → hydrophobic (insoluble in water).

3 Major Biological Classes — Fats (triglycerides) , phospholipids , and steroids .

Structure — 1 Glycerol (3-carbon alcohol with 3 –OH groups) covalently linked to 3 Fatty Acid chains via 3 ester linkages (formed by 3 dehydration synthesis reactions...

Thermal Insulation — Subcutaneous adipose tissue (blubber in whales/seals).

Elemental Makeup — Built from long hydrocarbon chains (C–H bonds) with very little oxygen (H:O ratio ≫ 2:1).

Fatty Acid Components — Carboxyl Group (–COOH): Acidic, polar head at one end.

Biological Functions — Long-Term Energy Storage: Dense, anhydrous (stores without heavy hydration water).

Mechanical Protection — Cushions vital internal organs (kidneys, heart).

§2. GENERAL PROPERTIES & FATTY ACID STRUCTURE

Hydrophobic C–H

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

General Properties & Fatty Acid Structure ➔ **Defining Characteristic** ➔ **Elemental Makeup** ➔ **3 Major Biological Classes** ➔ **Fatty Acid Components**

- **Defining Characteristic:** Grouped together NOT by repeating monomer polymers, but because they are **mostly nonpolar → hydrophobic** (insoluble in water).
- **Elemental Makeup:** Built from long hydrocarbon chains (**C–H** bonds) with very little oxygen (H:O ratio ≫ 2:1).
- **3 Major Biological Classes:** **Fats (triglycerides)**, **phospholipids**, and **steroids**.
- **Fatty Acid Components:**
  - **Carboxyl Group (–COOH):** Acidic, polar head at one end.
  - **Hydrocarbon Tail ($–(CH_2)_n–CH_3$):** Long nonpolar chain (typically 14–22 carbons); often makes lipids **amphipathic** (having both hydrophilic and hydrophobic regions).

[CORE AXIOM / REMEMBER] **Lipids = "water-fearing molecules"** (nonpolar **C–H** chains cannot form hydrogen bonds with water and are excluded by water's H-bond network).

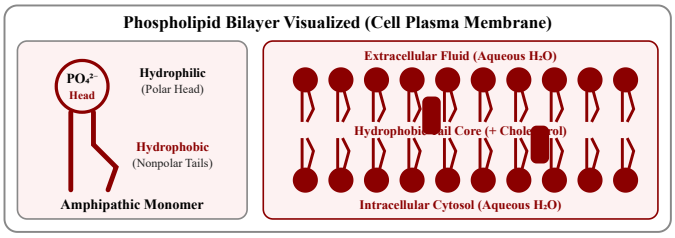
§5. PHOSPHOLIPIDS & BILAYER SELF-ASSEMBLY

Amphipathic

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Phospholipids & Bilayer Self-Assembly ➔ **Phospholipid Structure** ➔ **Amphipathic Nature**

- **Phospholipid Structure:** **1 Glycerol + 2 Fatty Acid tails + 1 negatively charged Phosphate group ($–PO_4^{2-}$)** (often linked to a polar choline/serine group).
- **Amphipathic Nature:**
  - **Hydrophilic Head** (glycerol + phosphate + polar group) = polar/charged → interacts with water.
  - **Hydrophobic Tails** (2 hydrocarbon fatty acid chains, often 1 saturated + 1 kinked unsaturated) = nonpolar → excluded from water.
- **Phospholipid Bilayer (Spontaneous Assembly in Water):**
  - Driven by hydrophobic interactions: **hydrophilic heads face outward** toward aqueous cytosol & extracellular fluid; **hydrophobic tails face inward** away from water.
  - **Emergent Membrane Properties:** **Selective permeability** (blocks ions/polar solutes, allows small nonpolar molecules), **lateral fluidity**, and **spontaneous self-healing**.



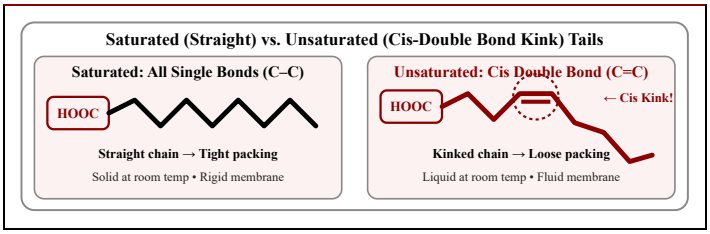
[CORE AXIOM / REMEMBER] **Bilayer = Sandwich: Bread** = hydrophilic heads facing water outside; **Filling** = hydrophobic fatty acid tails tucked inside.

[AP EXAM FRQ SCORING STRATEGY] Always link: **Amphipathic nature → spontaneous phospholipid bilayer formation in aqueous environments!**

§3. SATURATED VS. UNSATURATED FATTY ACIDS

Membrane Fluidity

FEATURE	SATURATED FATTY ACIDS	UNSATURATED FATTY ACIDS
C–C Bonding	**Only single C–C bonds** (saturated with max H atoms)	**≥1 cis double bond (** C=C **)** (fewer H atoms)
Chain Geometry	**Straight chains** → pack tightly via van der Waals forces	**Bent / kinked chains** at **cis C=C** → prevent tight packing
Room Temp State	**Solid** at room temp (e.g., butter, lard, animal fats)	**Liquid** at room temp (e.g., olive oil, plant & fish oils)
Membrane Effect	**Decreases fluidity** (more viscous / rigid membrane)	**Increases fluidity** (stays fluid at cold temperatures)



[EXAMPLE & BIOLOGICAL APPLICATION] **Butter** (saturated animal fat, solid) vs. **olive oil** (unsaturated plant oil, liquid); cold-water fish & winter wheat increase **unsaturated fatty acid %** in membranes so membranes don't solidify in freezing water!

[CORE AXIOM / REMEMBER] **Saturated = straight spaghetti** → packs tightly in the box.
Unsaturated = kinked spaghetti → loose packing with space between tails.

[AP EXAM FRQ SCORING STRATEGY] Always use the exact phrase: **"cis double bonds (** C=C **) introduce kinks in hydrocarbon tails, preventing tight packing and increasing membrane fluidity."**

§4. FATS (TRIGLYCERIDES / TRIACYLGLYCEROLS)

~9 kcal/g Energy

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE → MECHANISM → OUTCOME]**

Fats (Triglycerides / Triacylglycerols) → **Structure** → **Biological Functions** → **Energy Yield**

- **Structure:** **1 Glycerol** (3-carbon alcohol with 3 **–OH** groups) covalently linked to **3 Fatty Acid chains** via **3 ester linkages** (formed by 3 dehydration synthesis reactions releasing 3 **H_2O**).
- **Biological Functions:**
  - **Long-Term Energy Storage:** Dense, anhydrous (stores without heavy hydration water).
  - **Thermal Insulation:** Subcutaneous adipose tissue (blubber in whales/seals).
  - **Mechanical Protection:** Cushions vital internal organs (kidneys, heart).
- **Energy Yield:** **~9 kcal/g** (more than double the ~4 kcal/g of carbohydrates or proteins) due to abundant reduced, high-energy **C–H** bonds.

[CORE AXIOM / REMEMBER] Don't confuse energy density with speed: **Carbohydrates (4 kcal/g)** are for *quick, immediate ATP*; **Fats (9 kcal/g)** store *more than double the energy per gram* for long-term reserves!

§6. STEROIDS: CHOLESTEROL & STEROID SIGNALING

4 Fused Carbon Rings

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE → MECHANISM → OUTCOME]**

Steroids → **Steroid Structure** → **Cholesterol (Animal Cell Membrane Buffer)**

- **Steroid Structure:** Characterized by a rigid carbon skeleton of **four fused carbon rings** (3 six-membered rings + 1 five-membered ring) with varied functional groups attached.
- **Cholesterol (Animal Cell Membrane Buffer):**
  - Wedges between phospholipid tails in animal plasma membranes and acts as a **bidirectional temperature fluidity buffer**:
    - **At High Temperatures:** Restrains phospholipid movement → ↓ **decreases fluidity** (prevents melting/excessive permeability).
    - **At Low Temperatures:** Disrupts tight packing of saturated tails → ↑ **increases fluidity** (prevents freezing/solidification).
- **Steroid Hormones (Estrogen, Testosterone, Cortisol, Aldosterone):**
  - Because they are **nonpolar and lipid-soluble (hydrophobic)**, steroid hormones **diffuse directly across the phospholipid bilayer** without transport proteins.
  - Bind to **intracellular (cytoplasmic or nuclear) receptors** inside the target cell, forming a hormone-receptor complex that acts as a **transcription factor** to directly regulate gene expression!

[COMMON AP EXAM FRQ TRAP] **Confusing steroid vs. protein/peptide hormones:**
• **Steroid hormones** (estrogen, testosterone) are *nonpolar* → cross the plasma membrane and bind **intracellular receptors**.
• **Protein/peptide hormones** (insulin, epinephrine) are *polar/charged* → cannot cross the hydrophobic core and must bind **cell-surface membrane receptors**!

§7. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 1.5 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 1.5: Lipids: Fatty Acids, Triglycerides, Phospholipids & Steroids ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Lab Context: Macromolecule Indicator Assays (Benedict's for reducing sugars, Lugol's Iodine for starch, Biuret for peptide bonds) & Buffer Titration Curves. Independent Variable (IV): pH or temperature treatment; Dependent Variable (DV): absorbance/colorimetric change or structural stability; Negative Control: deionized H_2O. **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative & Graph Analysis: Stoichiometric monomer-to-polymer condensation: for n monomers joined in a linear chain, $(n - 1)$ covalent bonds form and $(n - 1)$ H_2O molecules are released. pH scale is logarithmic: $pH = -\log_{10}[H^+]$; each 1-unit decrease in pH represents a 10-fold increase in $[H^+]$. Caloric density: Lipids (~9 kcal/g) vs. Carbohydrates/Proteins (~4 kcal/g). **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Molecular Disruption Case Study: Missense mutation in the β-globin gene (Glu6Val in Sickle-Cell Disease) replaces a polar/charged glutamic acid R-group with a nonpolar hydrophobic valine on the protein exterior, driving abnormal quaternary hydrophobic aggregation into rigid fibers under deoxygenated conditions. Similarly, extreme fever (>40°C) or acidosis (pH < 7.0) disrupts weak hydrogen and ionic bonds in tertiary (3°) protein structure while leaving primary (1°) peptide bonds intact.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 1.5 to prove why the claim holds.

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